

Predetermined Change Control Plan (AI)

Grounded in: LumineticsCore™ / IDx-DR (De Novo DEN180001, 2018)

Basis: FDA PCCP final guidance (2024); device facts public.

Manufacturer	Digital Diagnostics (formerly IDx)
Device	LumineticsCore™ (formerly IDx-DR) — autonomous AI for diabetic retinopathy
Pathway / status	De Novo DEN180001 (2018) — first FDA-authorized autonomous diagnostic AI; later 510(k) on its own predicate (2021)
AI structure	Two algorithms: an image-quality AI and the diagnostic AI; outputs an autonomous mtmDR result
Public performance	Pivotal trial (900 patients); FDA performance thresholds 85% sensitivity / 82.5% specificity
Basis	PCCP per FDA final guidance (2024). Illustrative — shows how a PCCP could be structured for this device class.

1. Description of modifications

- Periodic re-training of the diagnostic algorithm on additional labeled fundus images (intended use unchanged).
- Bounded re-tuning of the operating threshold within pre-specified limits.
- Support for additional compatible fundus cameras after equivalence testing.

2. Modification protocol

Data management. Minimum dataset size and demographic representativeness defined; an independent, frozen test set with no leakage from training.

Re-training & evaluation. Acceptance gates pre-specified — e.g., maintain sensitivity/specificity at or above the cleared thresholds (85% / 82.5%) overall and per pre-defined subgroup; no degradation on the locked test set.

Release & rollback. Versioned release with documented validation and a rollback path.

3. Impact assessment

Modification	Risk	Control / verification
Re-training	Subgroup performance drift	Locked acceptance gates + subgroup analysis on frozen test set
Threshold re-tuning	Sensitivity loss	Bounded change; re-validate against thresholds
New camera	Image-domain shift	Equivalence testing before enablement

4. Transparency

Each release within the PCCP adds a version note to labeling (date, data summary, performance delta). Changes beyond this authorized envelope require a new marketing submission.

Sources (public)

- FDA De Novo DEN180001 (IDx-DR), 2018; Digital Diagnostics (LumineticsCore) public materials.
- Abràmoff et al., npj Digital Medicine (2018) — pivotal trial; thresholds 85% sensitivity / 82.5% specificity.
- FDA — Marketing Submission Recommendations for a PCCP for AI-Enabled Device Software Functions (final, 2024).